Abstract (will be removed for 10 page limit, only included for review): This research aims to develop social robots as agents for persuasive technologies that can autonomously interact with people and effect positive change on their behaviour. Currently, social robotics research predominantly takes place in the developed world. Also real world deployment of social robots is rare, with more than 75% of research done in a lab without considering real world impact. On the other hand, rural populations in developing countries make up almost half the world's population, but the influence and perception of social robots on these communities has barely been investigated which leaves a big scientific gap in the field of social robotics. In order to bridge this gap, I have identified hand hygiene as an initial case where social robots could create a very high impact in developing countries. In these countries diarrhoea and respiratory infections are the two biggest causes of child mortality, over 1,300 young children dying each day, or about 480,000 children a year. By developing an intervention such as hand washing with soap, child mortality can be reduced by 40%. In this project I propose a novel method of hand hygiene intervention by developing a social robot to encourage handwashing behaviour. Social robots can, for example, encourage, remind, educate, give real-time feedback, observe and monitor hand washing behaviour. I will apply a user-centered and interdisciplinary approach to develop a social robot and carry out empirical studies in five rural schools in India. Using the social robot I will analyse which verbal, non-verbal strategies from the social robot are most effective in influencing handwashing behaviour and how children respond to the robot. The outcomes of this research will contribute towards informing the design of future social robots, both for the challenges of the developing world and also broader applications for affecting positive behaviour change.

1 Excellence

1.1 Quality and credibility of the research/innovation project; level of novelty, appropriate consideration of inter/multidisciplinary and gender aspects

• **Introduction:** ENDEAVOUR aims to develop social robots as agents for persuasive technologies that can interact with people and effect positive change on their behaviour. Since most of the social robotics research is carried out in developed countries, my research aims to **(i)** fill the **scientific gap** in our understanding of how people in developing countries from rural backgrounds interact with social robots and **(ii)** develop the interaction strategies for a social robot to effect **positive change** on their behaviour. This research will be carried in context to a very important issue in public health in developing countries, hand hygiene compliance. Handwashing with soap is one of the most inexpensive and effective public health interventions which could **save** more than a million lives a year from diarrhoea and **prevent** respiratory infections by approximately 40%,[1] the 2 biggest causes of child mortality in developing countries.[2] Existing practices for influencing handwashing behaviour in developing countries face three major challenges **(i)** they are very resource intensive, **(ii)** complex to measure/monitor compliance at scale and **(iii)** they rarely are sustainable over long-term.[3] Hence there is need for an **interdisciplinary and innovative approach** for influencing sustainable handwashing behaviour. However, **none** of the previous hand hygiene interventions in developing countries have considered the use of interactive technology such as social robots. **In this project, I propose a novel method of hand hygiene intervention by developing a social robot to encourage handwashing behaviour and measure its impact by systematic empirical studies**. The key research questions to be addressed in this project are **(i)** what interaction design strategies (verbal and non-verbal) are the most effective in influencing handwashing behaviour? and **(ii)** how can social robots achieve sustainable handwashing behaviour change? The research methodology will involve: **(i)** an interdisciplinary approach combining behavioural science, gamification principles and multimodal affective feedback strategies from the social robot; **(ii)** a user-centered iterative design approach within the target population; **(iii)** empirical studies to investigate the influence of a social robot on handwashing behaviour.

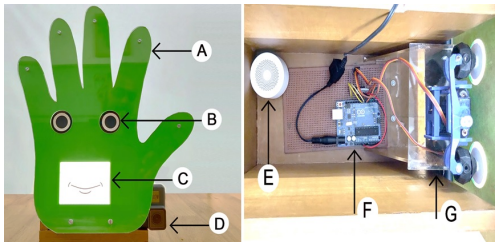
Project rationale: Social robotics research is mainly carried out in the developed world where people build expectations about robots due to their exposure to media and their interaction with modern technology. Rural populations in developing countries have limited exposure and access to different technologies due to their geographically remote and reduced economic background. Social robots could potentially create a positive impact in their lives but they have **rarely** been tested with people from rural backgrounds in developing countries. Previous research in developed countries has shown, robots in education could motivate children to learn better.[4] Handwashing education in schools are an effective way of reaching children and teaching them the habit of

[1] Dean T Jamison et al. The World Bank, 2006. [2] Tamer Rabie and Valerie Curtis. In: *Tropical medicine & international health* 11.3 (2006), pp. 258–267. [3] Emmy De Buck et al. In: *Campbell Systematic Reviews* 7 (2017), pp. 1–447. [4] Goren Gordon et al. In: *Thirtieth AAAI Conference on Artificial Intelligence*. 2016.

handwashing at a young age.[5] ENDEAVOUR will first target India, which has the world's largest rural population of **850 million**, with only **15%** practising handwashing with soap, and then seek to diversify to other countries after this fellowship. The annual net costs to India from not washing hands are estimated at US **$ 23 billion**, so a national handwashing program in India could provide an estimated **92-fold return** on investment.[6] ENDEAVOUR will carry out studies in 5 schools in 3 different states in India reaching out to approximately 1,000 children, aged 5-12 years. The outcomes of ENDEAVOUR will provide original understanding about how rural children interact with robots and interaction design strategies required by social robots to effect a **positive impact** in their health and well being.

State-of-the-art: In developing countries hand hygiene interventions mostly focus on educating people about health, germs and disease often using tools such as games, radio/TV campaigns, books, songs, posters, leaflets and charts.[7] Such methods can be very resource intensive and cumbersome to monitor at scale, measurements are unreliable due to self-reporting, and these approaches **rarely** result in positive, sustained behaviour change.[8] In developed countries, hand hygiene interventions have investigated the use of electronic message boards,[9] visual[10] and audio[11] prompts as a reminder to perform hand hygiene,[12] emoticon-based feedback,[13] and vision-based system for tracking people's activity in hospitals[14] with some degree of success. The state-of-the-art technology developed by SureWash (www.surewash.com) uses computer vision technology for hand hygiene training to hospital staff, costs about £30k, it is bulky, works only in indoor environments and has **limited** interaction capabilities. Summing up, most technology based hand hygiene interventions are expensive, they have been performed in developed countries in indoor environments and **none** have explored the use of social robots. In regards to social robots as persuasive applications, use of verbal[15] and non-verbal behaviour such as eye gaze[16] (i.e. looking at a participants' face), has shown positive results. However, whether the persuasive effects of the robot's verbal and non-verbal cues will influence decision making in real-world scenario such as handwashing is still **undiscovered**.

Proof-of-concept: Given the ambition and the risks in this research vision, I conducted an independent research pilot project in collaboration with Amrita University (AU) in India. I led a team of 4 researchers and developed a proof-of-concept a low-cost ($\approx$ US $ 100) social robot which was remotely controlled (not autonomous) in a rural village school in India (see fig. 1). The short two day pilot intervention with the social robot showed encouraging results: a staggering 40% increased level of hand washing and with a better handwashing technique[17] with instruction based verbal messages and eye movement from the robot. However, further research is needed to leverage this proof-of-concept and develop the technology further.



A: Robot Face,
B: Eye movement,
C: Graphical screen,
D: Front camera,
E: Speaker,
F: Microcontroller,
G: Eye Mechanism

Figure 1: Image Left: Prototype Robot Design; Image Right: The wall mounted social robot near the water tap encouraged around 100 children to wash their hands via a poster on the wall,[18] March 2019.

Objectives: The ultimate goal for ENDEAVOUR is to establish a solid empirical foundation to inform interaction strategies required for the social robot to influence the target subjects into adopting the decision to perform handwashing behaviour. The main objectives of this research are: **(1)** to evaluate if gamification and affective feedback are effective to influence handwashing behaviour with children in rural schools (Work Packages (WPs) 1-2-3); **(2)** to refine the robot design and interaction strategy following the evaluation by an informed design process (WP4); **(3)** to conduct field studies to measure the influence, perception and sus-

[5] Robert Dreibelbis et al. In: *International journal of environmental research and public health* (2016). [6] Joy Townsend, Katie Greenland, and Val Curtis. In: *Tropical medicine & international health* 22.1 (2017), pp. 74–81. [7] Adam Biran et al. In: *The Lancet Global Health* 2.3 (2014), e145–e154. [8] Val Curtis et al. In: *The Lancet infectious diseases* 11.4 (2011), pp. 312–321. [9] Gaby Judah et al. In: *American Journal of Public Health* 99.S2 (2009), S405–S411. [10] SJ Storey et al. In: *Journal of Hospital Infection* 88.2 (2014), pp. 84–88. [11] Dale A Fisher et al. In: *Infection Control & Hospital Epidemiology* 34.9 (2013), pp. 919–928. [12] Arjun K Venkatesh et al. In: *American journal of infection control* 36.3 (2008), pp. 199–205. [13] Susanne Gaube et al. In: *PloS one* 13.5 (2018). [14] Albert Haque et al. In: *Proceedings of Machine Learning for Healthcare* (2017). [15] Jaap Ham and Cees JH Midden. In: *ICSR* 6.2 (2014), pp. 163–171. [16] Kazuhiko Shinozawa et al. In: *IROS*. vol. 2. IEEE. 2004, pp. 1908–1913. [17] Amol Deshmukh et al. In: *RO-MAN*. IEEE. 2019.

tainability of the social robot (WPs 5-6-7); **(4)** to evaluate feasibility on automatic perception of behaviour cues from video data gathered from field studies and provide recommendations on use of social robots in technology based interventions (WPs 8-9). I aim to carry out interventions using the social robot in 5 schools overall (target reach approximately **1,000** children, aged 5-12 years) across 3 states in India during the research period.

Overview of the action: I will develop and engineer a **low-cost** ($\approx$ US $ 250) social robot to create a step-change in the efficacy of handwashing with soap. My proposed approach has a potential to increase hand hygiene compliance, which in return could **reduce** germ transmission, save public health costs and decrease infection-related absenteeism in schools. Also **contribute** towards meeting UN's sustainable development goals targets around sanitation, hygiene, education, nutrition and child survival in developing countries. The outcomes of this project will also provide **new and original** understanding of human behaviour towards technology based interventions in the context of handwashing using social robots which will **benefit** the scientific community. I will initially conduct studies in 5 rural schools in India.

• **Research methodology and approach:** The approach will focus on investigating the use of gamification which has shown positive results in technology based education[19] and health related contexts.[20] Gamification is the process of including game like elements that could motivate and influence human behaviour to enhance learning. Currently there is **no evidence** to suggest that gamification strategies are effective to increase hand hygiene compliance for children from rural backgrounds in developing countries using social robots. My previous work with affective feedback has shown promising results with social robots as tutors.[21] However, more empirical research is required to determine whether both extrinsic and intrinsic motivation to wash hands can be influenced by gamification combined with affective feedback strategies. ENDEAVOUR will follow a user-centered iterative methodology to evaluate the interaction design strategy necessary to address the scientific challenges in the context of handwashing by means of empirical studies in 5 rural schools in India.

Develop an initial system for the first evaluation for Objective 1- The initial interaction system will be developed on the robot used during pilot test (fig. 1). Previous handwashing interventions have shown the use of emotional drivers significantly increased hand washing, for example disgust (the desire to avoid and remove contamination) using an animated film/video.[22] However, the use of emotional drivers using affective feedback by means of social robots in hand hygiene context is **unexplored**. For affective expressiveness, the social robot will also have eye movement to track users and display emotions e.g. happy, sadness, disgust. A screen using a mobile phone to display graphical mouth to display smiles, frowns and a loud speaker for sounds and speech synthesis. Gamification strategies will include for example, if the children wash their hands effectively, following the 6 steps of the WHO hand hygiene technique (hand movement)[23] shown by a poster on the wall. A a time based graphics in relation to the handwashing steps completed will be shown on the screen (e.g. balloon bursts or a rocket fires). Also verbal and non-verbal feedback strategies using social reinforcement[24] will also be used (e.g., the robot expressing social praise by saying "You are doing a great job, keep it up" while showing a happy expression and sounds). My supervisor Dr Mary Ellen Foster from UGLA an expert on verbal and non-verbal communication with social robots will guide me on the interaction design.[25]

Field study 1: Evaluation of interaction and design strategies for Objective 1- A field study will be conducted by using Wizard of Oz (WoZ - a person remotely operating a robot's behaviour) which is an important technique in social robotics research to get initial impressions about interaction designs.[26] The aim of the study will be to investigate the following: **(1)** how gamification combined with affective feedback strategies (emotions) expressed through verbal and non-verbal behaviour from the robot alters its persuasiveness and how it may be perceived by participants; **(2)** what important perceptual cues that need to be recognised and handled for an autonomous system. Evaluating these characteristics will inform the design of interaction strategy and the features required for an autonomous system. A parallel-design, randomised cluster experiment will be performed near the handwashing station in 2 rural schools. A five day intervention with the robot will be conducted to compare two interaction styles, **(i)** social robot provides step-by-step instructions to wash hands; **(ii)** social robot with gamified and affective responses. **Measurements:-** Mixed research methods will be used i.e. qualitative and quantitative data will be collected using video recordings of the interaction and questionnaires. The video data will be analysed to

[19] Sara Kiesler et al. In: *Academic exchange quarterly* 15.2 (2011), pp. 1–5. [20] Pedro Pereira et al. In: *International conference of design, user experience, and usability*. Springer. 2014, pp. 742–753; Jeffrey Clark, Joel Reynolds, and Philip G Crandall. In: *Food Control* 107 (2020), p. 106772. [21] Mary Ellen Foster et al. In: *AAMAS*. 2015, pp. 1711–1712. [22] Biran, see n. 7; Renata Porzig-Drummond et al. In: *Social science medicine* 68 (Feb. 2009), pp. 1006–12. DOI: 10.1016/j.socscimed.2009.01.013. [23] https://www.who.int/gpsc/5may/How_To_HandRub_Poster.pdf. 2009. [24] Ham and Midden, see n. 15. [25] Foster et al., see n. 21. [26] Laurel D Riek. In: *Journal of Human-Robot Interaction* 1.1 (2012), pp. 119–136.

determine behaviour cues and social signals from the users such as proxemics (i.e. coming close to the robot), facial expressions (i.e., smiles), oculesic behaviour i.e. eye contact (frequency, duration), speech utterances, and body movements (gestures, head pose). Analyse the correlations between children's behaviour in response to robot's verbal and non-verbal behaviour. Annotations will be performed using BORIS, an event logging software for video/audio coding.[27] Videos of participants washing hands will also be recorded from a top camera. In addition, qualitative feedback will be obtained from focus groups using questionnaires and semi-structured interviews to get detailed perception about the design features of the robot. To make it easier for children to respond to the questions, a Smileymeter, a visual likert scale[28] will be used. The questionnaires will capture perceptions about appearance, verbal and non-verbal behaviours of the robot. Prof. Bhavani Rao (AU supervisor), an expert in rural interventions methodologies will guide me during this phase.

Refine design and interaction strategies for the social robot for Objective 2- Development of the refined system informed by the findings from field study 1, this involves two main components, refinement of: **(1)** physical design and **(2)** synthesis (verbal, non-verbal interaction). **(1) Physical design:** The appearance of the robot will be refined following feedback from focus group in first study and the new design will be fabricated using 3D printing facilities available at AU's maker space on campus in India. The robot will be powered using cost-effective computers such as Raspberry Pi and smart phones for screen on the robot to display graphical/educational contents relevant to handwashing. **(2) Synthesis:** I will incorporate the findings from field study 1 into interaction strategies for the social robot. Both verbal and non-verbal strategies such as using speech, sounds, eye-tracking[29] and graphical mouth using the screen on the robot will be developed. The verbal feedback will be delivered in the local regional language the children can understand. Child-like and gender neutral voices are most effective in child-robot interaction so we will incorporate that in our speech design.[30] As there is lack of text-to-speech synthesis systems for Indian regional languages, human voice recordings will be manipulated by shifting the pitch of the sound files to resemble that of a child whose gender is not apparent in the voice to avoid gender and age perception bias.

Field study 2: Measure the influence and perception of the robot for Objective 3- The aim of this field study is to investigate: **(1)** influence, **(2)** learning transfer, and **(3)** sustainability of the intervention. Field studies will be carried out (WoZ technique[31]) in 3 rural schools across different states in India (Kerala, Maharashtra and Uttarakhand), also to capture the cultural diversity in regards to perception and influence of the social robot. Staggered interventions will be carried out with **stepped-wedge design** methodology.[32] This methodology helps to **mitigate novelty effects** imposed by exposure of the intervention due to repeated interactions. In the initial stage to investigate the influence of the social robot, a 2 week (10 days) intervention will be carried in rural schools. Each study will be divided into 3 phases, **(i)** pre-robot intervention (baseline, no robot, 2 days), **(ii)** robot intervention, the robot will be placed at the handwashing station (6 days), and **(iii)** post-robot when the robot is removed (2 days). In all 3 phases handwashing video data will collected by placing an overhead (capturing handwashing) and front (capturing interaction) cameras near the handwashing station and pre-post questionnaires will be used as described in field study 1.

Analysis: (1) The influence of the social robot will be analysed by measuring verbal and non-verbal behaviours that positively correlate with desirable action from the participant, e.g. responding to the instructions from the robot and their reaction time. **(2)** Learning transfer will be measured by analysing the handwashing behaviour by manually annotating each handwashing session using BORIS software. Objective evaluation on the quality of handwashing will be measured by how many steps the children carried out of the 6 steps of handwashing[33] and the duration and frequency of handwashing during critical moments such as before meals and after use of toilet. **(3)** In the final stage, observational data near handwashing stations from schools (2 days) will be collected without the robot 6 months after the initial study to measure if the intervention was **sustainable**. Effort will be made to collect data from schools to measure the infection-related absenteeism numbers (no personal information) before and after field studies to analyse the overall effect and sustainability of the intervention. The synthesis of the results will also benefit from taxonomy of specific techniques to change behaviour.[34]

Analysis of data for Objective 4- The eventual ambition is to have system that works autonomously. The technical challenge is to recognise hand gestures and behaviour cues in outdoor in the light sensitive and unstruc-

[27] Olivier Friard and Marco Gamba. In: *Methods in Ecology and Evolution* 7.11 (2016), pp. 1325–1330. [28] Janet C Read and Stuart MacFarlane. In: *Conference on Interaction design and children*. ACM. 2006, pp. 81–88. [29] Ham and Midden, see n. 15. [30] Anara Sandygulova and Gregory MP O'Hare. In: *ICSR*. Springer. 2015, pp. 594–602. [31] Riek, see n. 26. [32] Karla Hemming et al. In: *Bmj* 350 (2015), h391. [33] https://www.who.int/gpsc/5may/How_To_HandRub_Poster.pdf, see n. 23. [34] Susan Michie et al. In: *Annals of behavioral medicine* 46.1 (2013), pp. 81–95.

tured environment due to diverse physical setups of handwashing stations in different schools. To determine the quality of handwashing, motion detection using background subtraction algorithm within a region tightly cropped into the area over the water tap will be analysed. Evaluation will be carried out using the video data gathered from field studies 1 and 2 to analyse the feasibility for training machine learning algorithms and facial analysis software to assess recognition of behaviour cues (e.g. body orientation, smile detector, head pose using k-means clustering algorithm) and quality of handwashing, for example the 6 steps of the WHO hand hygiene technique.[35] Meta analysis of the results from all field studies will be consolidated. The main outcome from the meta analysis will be to **(i)** provide **recommendations** on interaction designs for verbal and non-verbal behaviours using social robots with rural populations in developing countries; **(ii)** understanding about how people rationalise their behaviour and be **useful** in designing appropriate interaction strategies using social robots as agents for persuasive technologies.

• **Novelty:** Existing methods for hand hygiene promotion is achieved by means of, hygiene education, psycho-social theory, community-based working, social marketing, incentives or advocacy. These methods rarely result in sustaining behaviour change due to **lack** of interactive engagement with target users.[36] In regards to behavioural science it is known that people change their behaviour when they know they are being watched, also known as *"Hawthorne Effect"*.[37] Building on this premise, ENDEAVOUR proposes an **unprecedented** and **novel** approach to hand hygiene promotion and education by developing social robots as tools for persuasive technologies that can interact with people and could be effective to change their behaviour. The physical presence of a robot could impact human perception and influence behaviour as humans have a tendency to anthropomorphize social robots.[38] ENDEAVOUR will extend the scope of social robotics research into real-world high impact scenarios which is **extremely rare** in this field.

• **Interdisciplinary aspects:** ENDEAVOUR is highly interdisciplinary, and aims at filling the gap between current research practices in handwashing interventions such as: hygiene messaging, psycho-social theory, human factors and behavioural science embodied into a physical agent. I will investigate the effectiveness of the social robot in influencing behaviour change pertaining to hand hygiene by qualitative and quantitative research methodologies. The outcomes of ENDEAVOUR will provide both AU and UGLA with the evidence of the effectiveness of interdisciplinary approaches and this will help them to push their research forward.

• **Gender dimension:** Previous research suggests the importance of gender of the health hygiene promoter was a factor that could influence program effectiveness, for example women would not ask specific sensitive questions, such as birth control or personal hygiene, to a male health promoter.[39] Other studies have shown that in public health interventions, hygiene messages were perceived differently by men and women participants.[40] Gender dimensions will be given **special attention** by ENDEAVOUR. A gender neutral voice for the social robot will incorporated in order to mitigate gender perception biases. During studies we will ensure that male/female subjects are evenly balanced in numbers. I will also ensure we have female researchers in our team while conducting field studies.

1.2 Quality and appropriateness of the training and of the two way transfer of knowledge between the researcher and the host

Transfer of knowledge from the host institutions to the applicant: At AU, I will be joining the team at AMMACHI Labs, which is an academic and research centre at Amrita University that brings an interdisciplinary approach to addressing societal challenges. AMMACHI Labs research team has rich experience in carrying out field studies and technology deployment with large populations of rural participants in India and Africa. I will learn the design of qualitative and quantitative methodologies for carrying out field studies in rural villages which is novel in social robotics research. I also expect to develop novel qualitative and quantitative methodologies for social robotics research in developing countries working with expert researchers at AU.

Training on transferable skills: At **AU**, I will undertake **(1)** Research Methodology course, **(2)** Training Program on Research for Societal Good Through Social Responsibility, aligned with achieving the 169 targets leading to 17 United Nations sustainability development goals, **(3)** Defining and Refining - Research Methodology Workshop on First Steps in Research, and **(4)** Workshop on sustainable development solutions and **(5)** AU's Live-in-Lab program which is a multidisciplinary experiential learning initiative that facilitates researchers to engage with intervention development, and deployment of sustainable solutions for current challenges faced by rural commu-

[35] https://www.who.int/gpsc/5may/How_To_HandRub_Poster.pdf, see n. 23. [36] De Buck, see n. 3. [37] Henry A Landsberger. In: 3.2 (1960). [38] Cynthia Breazeal. In: *Annual international conference of the IEEE engineering in medicine and biology society*. 2011, pp. 5368–5371. [39] De Buck, see n. 3. [40] Judah et al., see n. 9.

nities in India. This knowledge and experience will help me to envisage potential social robotics applications that can serve these communities. At **UGLA**, I will undertake: **(1)** Research Integrity course, **(2)** Research Staff Conference, **(3)** Grant Writing Workshop, **(4)** Managing successful research projects workshop, **(5)** Career planning workshop, **(6)** Managing Research Teams, **(7)** Knowledge Exchange Support, **(8)** Glasgow Crucible a leadership and development programme designed to support early career researchers, and **(9)** Postdocs to Innovators (p2i) a European collaboration providing support and training for postdoc entrepreneurs. Workshops 1-7 are one day workshops and 8, 9 are three day resident workshops which has an application process.

Transfer of knowledge from the applicant to the host institutions: At **AU**, **(1)** I will engage with the Robotics Club in the department of Mechanical Engineering at AU, give a talk at the department about social robotics and the technical challenges of deploying robots in social environments, **(2)** I will run a 4 one-day seminar series on interaction design and data analysis, research methodologies for social robotics, open to students and researchers, **(3)** Through 2 seminars open to students and researchers, I will share the knowledge gathered through ENDEAVOUR about how children rationalise their behaviour in response to the robot's verbal and non-verbal behaviour which will expand the expertise on technology based interventions at AU and UGLA. **(4)** In addition to this, I shall also engage with the rural beneficiaries through awareness campaigns on hygiene during village intervention programs conducted by AU.

Transfer of knowledge from AU to UGLA: At **UGLA**, I will run 3 seminars on skills acquired from AU, focusing on aspects relevant to research methodologies, quantitative and qualitative research approaches in developing countries dealing with children, design of public health technology based interventions, and challenges of deploying technology based interventions in real-world environments. **(1)** I will share the results of ENDEAVOUR by conducting a talk at UGLA's social robotics group which will involve attendees from School of computing and Psychology, **(2)** Seminar at Glasgow Centre for International Development (GCID) in UGLA, which attracts researchers from different disciplines. **(3)** Seminar at Institute of Health and Wellbeing at UGLA. **(4)** Organise an international annual workshop on the topic "Social robots and artificial agents for developing countries: Challenges and opportunities" at the RO-MAN conference[41] with collaborators from AU. **(5)** Run workshops with primary/secondary school pupils using the social robot to increase awareness handwashing with soap through interactive games with the social robot. I am a STEM ambassador (Science Connects), who have links with all of the primary and secondary schools in the West of Scotland. UGLA currently does not yet have engagement with technology based interventions using social robots in developing countries. I also envisage ENDEAVOUR will open excellent research **opportunities** for further collaborations between other groups at UGLA and AU. The knowledge I will acquire at AU in regards to real-world deployment and research methodologies will **add value** to the social robotics group at UGLA.

1.3 Quality of the supervision and of the integration in the team/institution

Qualifications and experience of the supervisors: At **UGLA, Dr Mary Ellen Foster** is a Senior Lecturer. Her primary research interests are social robotics, embodied conversational agents, natural language generation, and conversational interaction. She is the **coordinator** of the MuMMER project a European Horizon 2020 project (7 partners) which is deploying a socially intelligent humanoid robot in a public shopping mall. Dr Foster has managed two postdoctoral researchers at UGLA, and as coordinator has also been responsible for a team of approximately **20** researchers from across the project partners. She is currently supervising five PhD students, including two through the UGLA's Centre for Doctoral Training in Socially Intelligent Artificial Agents, and has also supervised **17** Honours students (two **prize-winning** dissertations) and 13 MSc students over four years in Glasgow. She has published over 100 conference and journal articles in areas including social robotics, artificial intelligence, conversational interaction, and natural language generation (1528 citations, h-index of 24, i10-index 42). She is a co-director of the University of Glasgow Centre for Doctoral Training in Socially Intelligent Artificial Agents (SOCIAL). Supervisors at UGLA undergo a course of initial supervisor training immediately upon having supervision duties conferred upon them and this training is refreshed in 5-yearly cycles. I will also receive support from other Faculty members at UGLA: Prof. Alessandro Vinciarelli, an expert in Social Signal Processing, aimed at analysis, synthesis and automatic detection of nonverbal human-machine communication and Prof. Stephen Brewster, who has expertise on design of multimodal affective feedback using non-verbal behaviours in human-computer interaction domain. At **AU, Prof. Bhavani Rao** is the Director of AMMACHI Labs and she also serves as the **UNESCO Chair** in Women's Empowerment and Gender Equality at Amrita University. She heads the R&D

[41] https://robotics4good.github.io/socialrobots4development/

activities that focuses on Human-Computer Interaction (HCI), Technology Enhanced Learning and Robotics and Automation technologies for skill development, women empowerment and gender equality. The projects she investigates pan the domains of computerised vocational education and training for adults, 21st century skills education for children, community development through hygiene and sanitation, rural vulnerability mapping and disaster response and recovery. As the primary subjects of ENDEAVOUR will be children, I will have two mentors from AU, (1) Professor. Sidney Strauss, an expert in child development (3878 citations, h-index 25 and i10-index 43). He is Branco Weiss Chair for Research in Child Development and Education and (2) Dr. Meltem Akoyak-Yildiz at AU from pedagogic and education sciences. I will gain valuable knowledge on research methods for rural children that will further my aim to expand social robotics research in developing countries.

Hosting arrangements: UGLA hosted more than 30 Marie Curie Individual Fellowships between 2007 and 2014 and has 10 fellows from the 2014 IF call. 12 fellows came to Glasgow following the 2015 IF call and 9 fellows from 2016. 9 MSCA fellows came to Glasgow after the 2017 IF round. UGLA won 3 Fellowships in 2018. Since 2014, the Colleges have welcomed the following numbers of MSCA IF fellows: MVLS 19; Science and Engineering 21; Arts 3. AU has hosted numerous postdocs, visiting researchers from overseas collaborating with them. I will also have access to their maker space for building prototypes, office spaces, computing resources and library resources. AU are developing expertise in assembling interdisciplinary teams, making social scientists effectively working together rural communities. Prof. Bhavani Rao at AU will help me expand my network to other social scientists already involved in technology based interventions in rural communities which will help me to exchange ideas.

1.4 Potential of the researcher to reach or re-enforce professional maturity/independence during the fellowship

My key research expertise is to analyse how people interact with robots, building robotic systems that can interact effectively and influence human behaviour using user-centered design approaches. During my PhD I developed an autonomous system (autonomous navigation, user monitoring, vision perception) for a social mobile robot and investigated social verbal feedback strategies to influence user acceptance of a social robot. This was one of the **rare** long-term social robotics study carried out in a real-world workplace setting.[42] I have worked on multiple European Union projects involving social robots in schools, studying the effects of affective feedback in embodied robotic tutors to enhance the learning experience for children in schools,[43] and in public spaces, developing non-verbal behaviours for a humanoid robot and investigating human perception of those behaviours.[44] I have published 45 (20 as first author) peer-reviewed papers and journal articles internationally and reviewed numerous journal articles and conference papers. I am an **editorial board member** for the new open access Journal of Future Robot Life (JFRL). I was the **finance chair** for the IEEE RO-MAN conference in 2014 and a **special session chair** in 2019. I won the **best one-page statement** award on "What future research is required for personal robotics and why?" at the Towards Autonomous Robotic Systems conference, 2009. I built my first social robot during my Bachelor's in Electronics and **won national** level robotics competitions at Asia's largest model presentation contest, 2005. I received a **scholarship award** for Masters from Jacobs University Bremen M.Sc. in "Smart Systems", 2006. I have supervised 7 M.Sc student projects. I have always pursued independent research ideas that can cause a **real impact** on the ground in people's lives. Breaking the traditional research in my field, I took 2 months leave of absence and secured a researcher exchange funding of £3500 from Scottish Informatics and Computer Science Alliance (SICSA) to travel and collaborate with AU in India in 2017. I **led** a team from AU and **pioneered** the first human-robot interaction study in a developing country to study user perception and technology acceptance of a social robot in a rural village in collaboration with AU. The research resulted in 2 publications[45] and global media coverage such as, the BBC[46], IEEE Spectrum[47], The Telegraph[48], and a number of online media articles. The scientific excellence of UGLA and AU, combined with my own interdisciplinary skills with electronics and computer science, will provide me with the best career development path and an ideal platform for me to develop as a leader of social robotics research for developing countries. Before the start of ENDEAVOUR, under the guidance of my supervisors, I will prepare a career development plan to ensure a successful outcome of the project and to monitor progress in my researcher's

[42] Amol Deshmukh et al. In: *Frontiers in Robotics and AI* 5 (2018), p. 23. [43] Foster et al., see n. 21. [44] Amol Deshmukh et al. In: *Proceedings of the 6th International Conference on Human-Agent Interaction*. ACM. 2018, pp. 293–300. [45] Amol Deshmukh et al. In: *RO-MAN*. 2018, pp. 1–6. [46] https://www.bbc.co.uk/news/av/technology-43814332/the-robot-helping-to-deliver-water-in-rural-india
[47] https://tinyurl.com/ya4mwp5l [48] https://www.telegraph.co.uk/global-health/climate-and-people/robots-coming-fetch-water/

career. UGLA has been shortlisted for The Award for Outstanding Support for Early-Career Researchers for 4 separate initiatives since 2009. UGLA and AU offer a wide range of training courses and platform to provide researchers with exceptional support that enhance their potential to become leaders in their field.

2 Impact

2.1 Enhancing the future career prospects of the researcher after the fellowship

My long-term ambition is to develop low-cost high impact social robots as agents for positive behaviour change for developing countries. This prestigious Marie Curie fellowship will allow me to broaden my knowledge and skills and help to establish myself as a **world-leading** and a **pioneering** researcher in social robotics developing innovative solutions for rural communities in developing countries. **(1)** The interdisciplinary training I will receive, will allow me to expand my knowledge and skills in regards to quantitative and qualitative methodologies for conducting social robotics research with rural populations. **(2)** Social robotics research in developing countries is extremely **rare**, with this research I stand to become a **leading expert** in technology interventions using social robots in rural India. It will provide me with opportunities for career progression over the coming years, both in UGLA, AU and in other research institutions. **(3)** Research on verbal and non-verbal interaction strategies as persuasive agents in the context of handwashing will make me competitive with regard to technology based health interventions. **(4)** Since ENDEAVOUR will carry out field studies using tele-operation with final stage of the fellowship being dedicated towards evaluating technical challenges of an autonomous system. I envisage the latter could be my next project, possibly in collaboration with UGLA, AU in collaboration with other European partners. **(5)** AU are dedicated towards uplifting rural communities using technology based interventions, through ENDEAVOUR having connected to rural communities, other applications for social robots will be investigated by means of unstructured interviews with stakeholders. **(6)** The outcomes of ENDEAVOUR will contribute to social robotics research community, where most of the research is carried out in developed countries, in controlled lab environment and mostly with student participants. I envisage that my research will **inspire** other social robotics researchers to take up challenges to address the populations in the developing world. Through this fellowship I will attain professional maturity and be in good position to compete for tenure-track positions in Europe, e.g. in UK, the School of Computing Science at UGLA already has a strong track record in appointing Fellows to lectureship posts.

Relevance to Europe: In European schools for example a case study in Netherlands reported that, only 31% of children comply to hand hygiene guidelines. The economic burden of gastroenteritis and influenza-like illness attributable to children attending day care centres attendance, was estimated at €25 million and €72 million per year.[49] In UK for example, the total annual cost for treating children with infections in primary care was estimated to be £757 million. Institutions such as child day care centres and schools were identified as sources of infection. A hand hygiene interventions could bring estimated savings ranging from £106 to £387 per child per year[50] in UK. Health is the top priority among the Societal Challenges the EU. ENDEAVOUR if scaled up could potentially created a wider impact in regards to research in technology based health interventions using social robots in Europe.

2.2 Quality of the proposed measures to exploit and disseminate the project results

Both UGLA and AU will own the IP in the results arising from ENDEAVOUR, but I do not intend to protect them. This issue will be covered in the agreement between AU and UGLA. **(1)** I envisage that the research methods devised from ENDEAVOUR (e.g. new questionnaire designs for children, methodologies for collecting quantitative, qualitative data etc.) will be a valuable for many other researchers in social robotics. There is big community of researchers working in the domain of child robot interaction in developed countries. **(2)** The annotated video and image data generated from ENDEAVOUR (no identification/faces, hand movements only) will be valuable to train machine learning algorithms on hand gesture recognition for the wider community and will be made available on https://www.openml.org/ also through European Open Science Cloud. Annotated data on handwashing currently does on exist on open domains. **(3)** I will seek to form partnerships with policy makers and authorities to compliment the national campaign launched by Indian government "Swachh Bharat – Swachh Vidyalaya" (Clean India – Clean Schools) working to improve water, sanitation and hygiene (WASH) services and practices across India. **(4)** I will connect with NGOs and other stakeholders for example, public-private partnership with companies like Unilever (Lifebuoy) who are engaged in handwashing programmes worldwide[51]

[49] AH Elise van Beeck et al. In: *The European Journal of Public Health* 26.2 (2015), pp. 297–300. [50] Jayne Wilson, Dechao Wang, and Catherine Meads. University of Birmingham, Department of Public Health and Epidemiology, 2006. [51] Judah et al., see n. 9.

to pursue the exploitation of this research to wider scale. **(5)** I shall also pursue to take this research to other developing countries, I am an affiliate member of Sustainable Futures in Africa (SFA) network which spans across countries like Uganda, Botswana, Nigeria and Malawi. **(6)** At UGLA, Glasgow Centre for International Development (GCID), support staff members working with collaborators in low and middle income countries. Through GCID channels I shall seek to form new collaborations and exploit the results from ENDEAVOUR. **(7)** Meta analysis of the results from all field studies will be consolidated into 2 journal publications such as International Journal for Social robotics and Open Access journals (Lancet Global Health).

2.3 Quality of the proposed measures to communicate the project activities to different target audiences

The results from ENDEAVOUR will be disseminated through social media channels through AU and UGLA's Twitter account (95K followers), UGLA's Facebook (249.5K followers), press releases, local meetings, workshops in local schools in Glasgow. **(1)** Public engagement activities that AU participate in raising hand hygiene awareness in villages in India. **(2)** On the last after the study in each school, the social robot will be taken to each classroom to demonstrate the working of the robot for the 3rd and 4th grade students to encourage their interest in STEM disciplines. UGLA, events: **(3)** Science festivals in Edinburgh and **(4)** Glasgow attracting more than 50,000 visitors, and events such as, **(5)** Write a blog for Global Handwashing Day [52], a global advocacy day dedicated to increasing awareness and understanding about the importance of handwashing with soap. Previously I have given demonstrations in events like **(6)** SICSA Demofest [53] and **(7)** EXPLORATHON European Researchers' Night. These events provides a perfect opportunity for private and public sector organisations to experience the innovative computing research taking place in Scottish Universities and engage with public. I was speaker at the Pint of Science, "Robots, it makes sense!", Glasgow, 2017 [54]. At UGLA, **(8)** I will also participate in Glasgow Science Centre "Curiosity Live", activity. These public engagement activities will allow me to promote my research, also obtain feedback in Glasgow. **(9)** Work with the University's Widening Participation team to identify schools most in need e.g. areas of multiple deprivation according to the Scottish Index of Multiple Deprivation (SIMD) for public engagement activities for hand hygiene with the social robot. I will work with UGLA's and AU's Public Community Engagement Advisor to design an evaluation strategy for all my engagement activities, allowing me to monitor your success and impact.

3 Quality and Efficiency of the Implementation

3.1 Coherence and effectiveness of the work plan, including appropriateness of the allocation of tasks and resources

My plan for ENDEAVOUR is distributed into 9 workpackages (WPs), where the outcome of each WP informs the design strategy for the next WP. There is also the possibility for WPs to run in parallel to some extent as some activities can be carried out independently. I expect to publish the findings from the outcomes generated from WP3, WP6 and WP9 in relevant and leading conferences and journals. Legend: **WP** = Work package; **D** = Deliverable; **M** = Milestone. In the Gantt the horizontal bars show the duration of each WP.

WP1: Fieldwork starting phase (2 Months). **D1.1**. List of schools and initial discussions with school authorities in the selected villages. Ethics approval will be obtained. Start discussions with school authorities and researchers from AU who will assist with field studies. During this phase an initial interaction system with the planned approach will be developed for the robot, (fig. 1).

WP2: Field study 1 (1 Month). **D2.2** Study with initial system will be deployed in 2 rural schools. The study will be 5 days intervention in each school using the social robot. Focus group interviews, questionnaires, video data from the interactions will be collected. The expected number of group interviews, questionnaires will be 40 from each school. Expected video interaction (activity near the handwashing station) will be around 20-30 hours for both schools.

WP3: Data analysis (2 Months) **D3.1**. I will transcribe all interviews and then prepare, annotate videos, validate, and analyse WP2 data to refine the interaction and physical design strategy (WP4) for the social robot be used for the next field study (WP5).

WP4: Developing the system (3 Months) **D4.1**. WP3 results will be used to inform and develop the refined interaction design and build the new social robot hardware at maker space available at AU.

WP5: Field study 2 (4 Months). **D5.1** The refined system on the robot will be deployed in 3 rural schools (10

[52] https://globalhandwashing.org/global-handwashing-day/

[53] https://www.youtube.com/watch?v=dUdCS1emZ5s

[54] https://pintofscience.co.uk/event/robots-it-makes-sense

days in each school). Consent will be obtained from school authorities and children's guardians. Data collection methodology will be the same as mentioned in WP2. The expected number of group interviews, questionnaires will be 40 from each school. Expected video interaction, recording activity at the handwashing station will be around 50-60 hours for both schools. More time is given for this WP given the logistics of travel between locations and prior practical arrangements.

WP6: Data analysis and synthesis of results (5 Months) **D6.1**. Analysis of WP5 data will be performed which will include Interviews transcription, data annotation from videos and interviews responses (similar to process mentioned in WP3). To accelerate the video annotation process, system logs (when robot behaviours were triggered) correlated with video time-stamps will be analysed.

WP7: Perception data analysis (3 Months) **D7.1**. Evaluation of perception capabilities that could be developed using videos from handwashing, behaviour cues will be systematically assessed. Use of software tools for machine learning and vision processing (automatic affect detection) will be evaluated.

WP8 Measuring sustainability: (2 month) **D8.1** Data collection through AU from 3 schools mentioned in WP5 (six months after interventions in WP6). This data will include the frequency of handwashing behaviour obtained via (2 day manual observation), infection-related absenteeism numbers (no personal information), and interviews from school authorities to get feedback about the intervention. This data will be provided by researchers from AU who will visit the schools and I will analyse the data at UGLA.

WP9 Meta analysis: (2 months) **D9.1**, Results from WP3, WP6 and WP7 from all studies will be consolidated. Guidelines document for interventions using social robots in rural schools.

Milestones: **M1**: Career Development Plan drawn up; pre-fieldwork organisational phase concluded. **M2**: Field Study 1 completed. **M3**: Field Study 2 completed. **M4**: Data analysis on field study 2 completed. **M5**: Perception data analysis completed. **M6**: Data from 6 months after intervention collected. **M7**: Meta analysis completed.

	Outgoing Phase at Amrita University												Returning Phase at University of Glasgow											
Month	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
WP1																								
WP2																								
WP3																								
WP4																								
WP5																								
WP6																								
WP7																								
WP8																								
WP9																								
Deliverable		D1.1	D2.1		D3.1			D4.1				D5.1					D6.1			D7.1		D8.1		D9.1
Milestone			1					2				3					4			5		6		7
PM	CD	RM				PD				RM								PD	RM					
Training			1		2		3	4	TBC					9		TBC				8		TBC		
Dissemination						3			4		8	1						5	6			2	7	
Communication				1						2					9				3	8	4	6	7	5

Progress Monitoring (PM): CD = Career Development goal monitoring; PD = Performance and Development review; RM = Risk Management (see 3.2). For Dissemination (see 2.2) and Communication (see 2.2). Training activities (TBC) are activities that don't have assigned dates.

3.2 Appropriateness of the management structure and procedures, including risk management

UGLA has the overall responsibility for the administrative and financial aspects of ENDEAVOUR and the grant agreement preparation and partnership agreement phase. Its Research Support Office has extensive experience in managing EU projects and will give support to manage the grant and to communicate with the Research Executive Agency. This support will be aligned with the principles of the European Charter for Researchers. The EU Team at UGLA will also monitor project implementation, ensuring UGLA obligations under the Grant Agreement are met.

Financial Management: I will manage the budget under the guidance of Dr Mary Ellen Foster, in accordance with the terms and conditions of the Grant Agreement and following UGLA's financial guidance and regulations. A highly experienced and well-resourced EU Funding team will support Dr Mary Ellen Foster and me with all administrative, financial and legal aspects. The EU Funding Team will set up the project budget in line with the MSCA-GF unit cost categories. Expenditure will be closely monitored to ensure that only eligible project costs are met from the budget and that ENDEAVOUR can be successfully delivered.

Management of the project: Dr Mary Ellen Foster will be the scientific leader of the project, providing appropriate mentorship and supervision to me. Prior to starting the project, Dr Mary Ellen Foster and me will confirm

the objectives and goals of the project including specific tasks, milestones and deadlines (based on the Gantt chart provided). I will meet her once a week during UGLA phase and once every fortnight through Skype during outgoing phase to review and monitor progress. The supervisors will help me monitor the progress of my project, support me in the implementation of my career development plan, contribute to my publications and presentations, and help me in broadening my research network. UGLA operates a formal annual Performance and Development Review (PDR) as part of staff management. The aim of this is to reflect on performance during the previous year; achievements made; and to identify areas where progress can be made in the coming year, including the identification of any hurdles to be overcome in order to achieve these objectives.

Risk Management: Risks associated with ENDEAVOUR will routinely reviewed and updated throughout the project to reflect progress made in implementing the mitigation actions (RM See Gantt). **Risk 1** (real but properly mitigated): Challenges to find local rural schools who will collaborate, AU already has local branches in other states which will enhance our reach to new schools. Contingency plan: I am in touch with Public Health Foundation of India (PHFI), an independent public-private initiative whose vision is to strengthen India's public health also have connections with in rural schools, they have shown interest to collaborate. **Risk 2** (negligible): Propagation of delays on subsequent WPs. Contingency plan: I have months 23 and 24 before the end of the fellowship which are planned for the training, dissemination and publication writing but they could be used as a buffer period. **Risk 3** (negligible): Children/school authorities, refuse to collaborate with me. Contingency plan: The influential AU team will mitigate this risk. Also originally being brought up in India I am well aware about the culture and social structure in Indian rural environments which should allow me to conduct my work comfortably.

3.3 Appropriateness of the institutional environment (infrastructure)

UGLA and AU are the best institutions for me to accomplish the aims of ENDEAVOUR. At UGLA, I will work within the Glasgow Interactive Systems section (GIST) in the School of Computing Science, where much of the research is collaborative and interdisciplinary. GIST work closely with other groups in Computing Science as well as other schools, including Psychology and the Institute of Health and Wellbeing. My research will compliment School's research areas and will enhance its research profile, particularly growing the social robotics theme. At AU their team are at the forefront of technology based intervention research in rural India and they will provide me with the most comprehensive training on research methodologies. AU is actively working with over 57 rural schools, over 126 rural villages across 21 states in India where researchers from AU have built trust with rural administration and school authorities from their previous and ongoing work which will allow me to conduct this research comfortably.

4 CV of the Experienced Researcher

Dr Amol Deshmukh
http://www.dcs.gla.ac.uk/ amol/

Email:Amol.Deshmukh@glasgow.ac.uk
ORCID ID: **0000-0003-4942-3621**

Research Experience

University of Glasgow, Glasgow, UK *1/04/2016-current*
Research Associate

- Currently working with Psychology department on six months project to learn **new research methodologies** from social sciences perspective working with Prof. Stacy Marsella.
- MuMMer Project- European Union Horizon 2020: Overall research goal of developing a humanoid robot that can interact autonomously and naturally in the dynamic environments of a public shopping mall, providing an engaging and entertaining experience to the general public.
- Developed social **signal processing** components to convey personality dimensions, intentions using parameterised gestures, multi-modal non-verbal behaviour generation for a shopping mall robot.
- Implemented machine learning algorithms to recognise engagement/disengagement of users, infer users' social state interest and attention, conducted user studies to **analyse** user perception on non-verbal behaviours.

Heriot-Watt University, Edinburgh, UK *01/09/2008 - 31/032016*
Research Associate Worked on **3 European Union** FP7 projects.

- EMOTE, EU FP7, (01/12/2012-31/03/2016): Empathic robotic tutors for children in schools.
  - **Evaluated** interaction effects between children and a robotic tutor, such as movement, appearance, expressiveness and memory in a pedagogical setting.
  - Integrated **theoretical and pedagogical** work by project partners using **user-centred design** approach, conducted user studies with School children in Scotland.
  - Integration software modules (vision and perception), interaction modalities into a generic architectural framework.
  - **Led technical meetings** with project team members, produced test plans.
  - Conducted user studies with School children in Scotland.
- JAMES, EU FP7 (01/09/2012-30/12/2012): A robot bartender to socially interact with humans. Developed interaction behaviours for humanoid robot, analysed video data from user studies.
- LIREC, EU FP7 (01/09/2008-01/08/2012): Interactive, emotionally intelligent companion robots.
  - **Investigated** the service degradation caused by recharging and explored social verbal strategies used to influence users' expectations during recharge, **long-term deployment**.
  - Integrated system modules, developed competencies for robot e.g. human-aware navigation, user monitoring, vision perception, autonomous recharging.
  - Designed and engineered hardware (electronics, mechanical) for mobile social robot.
  - Developed **technical capabilities** for social mobile robot, work-package deliverables for EU, monthly reports to EU, collaborate with other partners in the project, led technical meetings with project partners.
- Designed and constructed the mechanical and electrical components of social robot platform.
- Supervised 7 M.Sc projects, demo presentations for University open days and public events.

Industry Experience

Bleetech Innovations, Mumbai, India *01/012018 - 31/06/2018*
Mentor/Technical advisor

- Provided **technical advise** and recommended user-centred design approaches for their innovative Blee Watch, a smart watch designed for the needs of the deaf people.

Fraunhofer IPA, Stuttgart, Germany 01/02/2008-31/07/2008 & 01/06/2007-31/08/2007

Robotics Engineer (Internship & Masters Project)

- Thesis: "An Intelligent Human Robot Interaction System for Entertainment Mobile Robot."
- Developed a system for an entertainment mobile robot used in public places such as museums, shopping malls and exhibitions (CeBIT, Hannover, Germany).
- System Architecture: **Behaviour based framework**, an intelligent mechanism that accomplishes a particular task using knowledge of previous interactions with humans (using **Reinforcement learning** approach), ability to customise the tasks performed by the robot using a graphical user interface.
- Monitor the performance of behaviours during interactions with humans and maintain performance scores for each behaviour, ability to customise the tasks performed by the robot.
- Implemented face detection, **person tracking**, human-Aware navigation, behaviour based control, monitoring and adaptations of interactions with users.
- Mobile camera robot control programming, KUKA arm control programming, workspace designing, GUI development, Mobile platform Control (C/C++, Java).

Cosmic Technologies, Pune, India 01/01/2006 - 30/06/2006

Software Engineer (Internship & Master Thesis)

- Developed online Toll Free Numbers Shopping & Management System (A.S.P/ I.I.S).

Leaps & Bounds, Pune, India 01/06/2002 - 31/01/2003

Software Engineer: **Single-handedly** developed 3 web portals, full software development life-cycle.

Education

Heriot-Watt University, United Kingdom *01/09/2008 - 01/12/2015*

Ph.D, "Human-Robot Interaction"

- Thesis: "An Investigation Of Service Degradation In Long-term Human-robot Interaction With a Particular Reference To Recharge Behaviour."
- Short Abstract: This work investigated service degradation in long-term interaction due to recharge behaviour of autonomous social mobile robots and proposed an approach to manage service degradation due to recharging behaviour of the robot using **social verbal strategies**.
- Technical Features: Human-aware navigation, path planning, vision perception, **user monitoring and tracking**, autonomous recharging, power management.

Jacobs University Bremen, Germany *01/08/2006 - 01/08/2008*

M.Sc. Computer Science, "Smart Systems" (**Scholarship Award**, GPA 1.00, A+)

- Courses: Advanced robotics, Motion planning, Autonomous systems, Speech signal processing, **Computer vision and perception**.
- Thesis: "An Intelligent Human Robot Interaction System for Entertainment Mobile Robot".

Pune University, India *01/07/2003 - 01/07/2006*

Master of Computer Application (M.C.A)

- Courses: Software Engineering, Network Programming, Artificial intelligence, Database Management, Object Oriented Programming C++/ Java, Statistical Computing
- Thesis: "Online Toll Free Numbers Shopping & Management System" (A.S.P/I.I.S).

Pune University, India *01/07/2000 - 01/07/2003*

Bachelor of Science, Electronics, (B.Sc.)

- Thesis: "Gyro the Humanoid robot: A self conceptualised prototype of a humanoid robot."

Research Funding

- Lord Kelvin Adam Smith (LKAS) Fellowship funding from UGLA **£100,000**, 2019.
- Travel grant of £3500 from SICSA - Scottish Informatics and Computer Science Alliance for postdoctoral and Early Career Researcher Exchange to collaborate with Amrita University in India, November 2017. (Independent research project under absence of leave).
- Contributed in defining and procuring robotics lab equipment for Heriot-Watt University, Edinburgh Centre for Robotics, 2014 (successful grant £6 million of UK government funding).

Prizes

- **Best one page statement** award on "What future research is required for personal robotics and why?", Towards Autonomous Robotic Systems Conference, Ulster, 2009.
- Master's project "Gyro the humanoid robot" - 3rd Prize at Techkriti at I.I.T Kanpur (**Asia's Largest** Model Presentation contest), 2nd Prize at Innovations, 2nd Prize at Robotsav, got immense media publicity in India, 2005.
- Student of the year award Masters in Computer Application, 2005.

Conference Organisation and Memberships

- **Special session chair** for IEEE International Symposium on Robot and Human Interactive Communication (RO-MAN 2019), New Delhi, India, **Finance chair** (RO-MAN 2014).
- **Main organiser** for inaugural workshop on "Social robots and artificial agents for developing countries: Challenges and opportunities" (RO-MAN 2019).
- **Editorial board member** of new open access Journal of Future Robot Life (JFRL). I have also been a reviewer for HRI, RO-MAN, Journal of behavioural robotics, ICSR.
- Affiliate member of Sustainable Futures in Africa (SFA) Network, Interdisciplinary network for researchers to build understanding, research, and practice in sustainability in Africa.

Research Expeditions

- I **led** a team of 4 researchers from Amrita University and developed a proof-of-concept a social robot to influence handwashing behaviour in a rural village school with 100 children, 2019.
- **First human-robot interaction** study in a developing country to study user perception and technology acceptance of a water hauling social robot in a rural village, I led a team of 4 researchers from Amrita University, under leave of absence from UGLA, India, 2017.

Dissemination Activities

- Independently led research on human-robot interaction in rural India published in more than 15 media articles globally. **IEEE Spectrum**, **BBC Click**, **The Telegraph**, Der Spiegel, March 2018.
- **Speaker** at Pint of science, "Robots, it makes sense!", Glasgow, 2017.
- Exhibitor at EXPLORATHON European Researchers' Night, "Pepper the robot" 2016.
- **Speaker** at Nesta event, London "Humanising the robot society", June 2014,
- The Scottish Informatics and Computer Science Alliance (SICSA), Demofest, November 2014.
- Live robot **demonstration** at Edinburgh International Science Festival, April 2012, "Let's Be Friends: Robot Companions".
- Video demonstration at CeBit, Hannover, Germany March 2012, "Exploring and Designing our future robot Companions".
- Featured in "Facial perception documentary", Royal Society Summer Exhibition, June 2011.
- Featured on **Euronews** (reach 400 million homes across 160 countries) "Futuris" Program on social robotics, March 2010

Invited Talks

- Seminar on "Human-robot Interaction: Challenges with User Studies", at Glasgow Interactive SysTems (GIST), 2018.
- "Educational Robots", at Amrita University, India, 2017.
- "Empathic Robotic Tutors", at Napier University, Edinburgh, UK, 2016.
- "Gyro the Humanoid Robot", presentations and talks at Schools and Management colleges, India, 2005.

Extra-Curricular

- Captain of Heriot-Watt University Cricket team, West of Scotland cricket club cricket team.
- Participated in 5 trekking expeditions in Himalayas as a group leader.
- Hobbies: Avid photographer, travelling and music.
- Languages: English, Hindi, British and Indian Sign language, German (intermediate).

Selected Publications

Published **45** peer-reviewed papers (**20 as first author**) internationally.

1. **Amol Deshmukh** et al., "Influencing Hand-washing Behaviour With a Social Robot: HRI Study With School Children in Rural India", *The 28th IEEE International Conference on Robot and Human Interactive Communication (RO-MAN), October, 2019, New Delhi, India.* [accepted]

2. Unnikrishnan R, **Amol Deshmukh** et al., "Design and Perception of a Social Robot to Promote Hand Washing among Children in a Rural Indian School", *The 28th IEEE International Conference on Robot and Human Interactive Communication (RO-MAN), October, 2019, New Delhi, India.* [accepted]

3. **Amol Deshmukh** et al., "Shaping Robot Gestures to Shape Users' Perception: The Effect of Amplitude and Speed on Godspeed Ratings", *In 6th International Conference on Human-Agent Interaction, December, 2018, Southampton, United Kingdom. ACM*

4. **Amol Deshmukh**,et al., "Technology Acceptance, Sociocultural Influence and Gender Perception of Robots: A Human Robot Interaction Study with Naive Users in Rural India", *The 27TH IEEE International Conference on Robot and Human Interactive Communication (RO-MAN), August, 2018, Nanjing, China.* [Independently led research]

5. **Amol Deshmukh** et al., "The More I Understand It, the Less I Like It: The Relationship between Understandability and Godspeed Scores for Robotic Gesture", *The 27TH IEEE International Conference on Robot and Human Interactive Communication (RO-MAN), August, 2018, Nanjing, China.*

6. **Amol Deshmukh** et al., "Social Impact of recharging activity in long-term HRI and verbal strategies to manage user expectations during recharge", *Frontiers in Robotics and AI- Humanoid Robotics Journal, 2018.*

7. **Amol Deshmukh** et al., "Modulating the Non-Verbal Social Signals of a Humanoid Robot", *19th ACM International Conference on Multimodal Interaction, November 2017, Glasgow, Scotland, United Kingdom.*

8. **Amol Deshmukh**, Mary Ellen Foster and Alessandro Vinciarelli, "Perceptual Control Theory for Engagement and Disengagement of Users in Public Spaces", *International Conference on Human-Robot Interaction, Vienna, Austria, March 2017.*

9. **Amol Deshmukh** et al., "How Expressiveness of a Robotic Tutor is Perceived by Children in a Learning Environment", *International Conference on Human-Robot Interaction, Christchurch, New Zealand, March 2016.*

10. **Amol Deshmukh** et al., "An Empathic Robotic Tutor in a Map Application", *Proceedings of the Tenth Annual ACM/IEEE International Conference on Human-Robot Interaction Extended Abstracts (Demo), 311-311, Portland, USA, March 2015.*

11. Mary Ellen Foster, **Amol Deshmukh** et al., "How does affective robot feedback influence learner experience in a real-world treasure hunt?", *In Proceedings of First International Workshop on Educational Robots (WONDER), 2015.*

12. Katrin Solveig Lohan, **Amol Deshmukh**, Ruth Aylett, "How Can a Robot Signal Its Incapability to Perform a Certain Task to Humans in an Acceptable Manner?", *In Proceedings of the 23rd IEEE International Symposium on Robot and Human Interactive Communication (IEEE RO-MAN 2014), Edinburgh, UK, 2014.*

13. **Amol Deshmukh** et al., "Towards empathic artificial tutors", *Proceedings of the 8th ACM/IEEE international conference on Human-robot interaction 2013, Tokyo, Japan.*

5 Capacity of the Participating Organisations

Participating organisations	Legal Entity Short Name	Country	Supervisor	Role of partner organisation
Beneficiary				
University of Glasgow	UGLA	UK	Dr Mary Ellen Foster	
Partner Organisation				
Amrita Vishwa Vidyapeetham	AU	India	Prof. Bhavani Rao	Hosting the outgoing phase

Beneficiary: University of Glasgow, UGLA, UK	
General Description	Founded in 1451, the University of Glasgow has a tradition of excellence as the UK's fourth oldest university. It is a member of the elite Russell Group of leading UK research universities, a founding member of Universitas21, and of the Guild of European Research-Intensive Universities. UoG has fostered the talents of **seven Nobel laureates**, Scotland's first female medical graduates, and includes among its alumni, some of the world's most renowned innovators, from scientist Lord Kelvin, economist Adam Smith, to the pioneer of television John Logie Baird. Ranked 67 in the world (QS World Rankings 2020), UoG is a **world-leading** research-intensive institution, attracting scholars from more than 130 countries worldwide each year. The School of Computing Science (SoCS) at UGLA is in the UK's 2014 independent research exercise are rated **top in Scotland** for research impact with 68% of impact judged world-leading and 32% internationally excellent. In the overall research ranking, SoCS was judged equal 16th amongst UK computer science departments, rising to 10th position on research volume with 84% of all research judged world-leading or internationally excellent. SoCS are **6th** in the UK on research intensity-weighted GPA rank order. SoCS is ranked 2nd in Scotland in The Complete University Guide 2019, joint 10th in the UK in THE World University Rankings by Subject 2018, and is in the top 100 in the QS World University Rankings by Subject 2018. Current research funding amounts to more than **£6M** in SoCS. The Social Robotics group at SoCS which is collaborative and **interdisciplinary**, working closely with other schools including Psychology and the Institute of Health and Wellbeing.
Academic organisation	Yes
Role and profile of key persons (supervisor)	**Dr Mary Ellen Foster** is a senior lecturer at SoCS at UGLA. Her expertise are in designing, implementing, and evaluating practical artificial systems that support embodied interaction with human users and **real-world deployment**. She is the coordinator of the MuMMER project European Horizon 2020 project and is also a co-investigator on the UK-funded project SoCoRo, which is developing a social robot for adults with autism. She was a work package leader on the European project JAMES, which developed and evaluated a socially intelligent robot bartender. Dr Foster has She has reviewed grant proposals for the European Commission as well as the Dutch, French, and Polish national research councils. She guest-edited a special issue of IEEE Transactions on Affective Computing. She is a Review Editor for Frontiers in Human-Robot Interactions, and is currently a senior programme committee member for the HRI 2020, ICMI 2019, EMNLP 2019, and IEEE RO-MAN 2019 conferences; she is an organiser of a workshop series on public-space human-robot interaction and also regularly reviews for journals and conferences in the areas of human-robot interaction, artificial intelligence, and computational linguistics. She has led or collaborated in the acquisition of approximately £20 million. She is an organiser of a workshop series on public-space human-robot interaction and also regularly reviews for journals and conferences in the areas of human-robot interaction, artificial intelligence, and computational linguistics.
Dept.	School of Computing Science
Key research facilities, Infrastructure and Equipment	I will be provided with a fully equipped working station (with PC, printer, free access to Internet, e-mail account, telephone) with an own desk located in the main University campus. I will also have free access to UGLA's libraries; free access to various academic courses; web-profile on UGLA website; access to online scientific journals; and venues for meetings/seminars/lectures. I will have access to AU's digital resources from UGLA campus.
Independent research premises?	Yes
Previous and current involvement in research and training programmes	UoG is currently participating in: 27 MSCA ITNs (from Horizon 2020), 42 MSCA IFs (from Horizon 2020), 5 MSCA RISE projects. The SoCS has recently been awarded a Centre for Doctoral Training (CDT) together with the Institute for Neuroscience and Psychology to train a cohort of 50 PhD students on Socially Intelligent Artificial Agents. I will have the opportunity to interact with PhD students from CDT and experts from School of Psychology.
Relevant publications and/or research/innovation products	(1) Mary Ellen Foster. "Natural language generation for social robotics: opportunities and challenges". In: Philosophical Transactions of the Royal Society B: Biological Sciences 374.1771 (Mar. 2019), p. 20180027. (2) Bart Craenen, Amol Deshmukh, Mary Ellen Foster, and Alessandro Vinciarelli. "Shaping Gestures to Shape Personalities: The Relationship Between Gesture Parameters, Atributed Personality Traits and Godspeed Scores". In: Proceedings of the 27th IEEE International Symposium on Robot and Human Interactive Communication (RO-MAN), 2018. (3) Mary Ellen Foster, Andre Gaschler, and Manuel Giuliani. "Automatically Classifying User Engagement for Dynamic Multi-party Human–Robot Interaction". In: International Journal of Social Robotics 9.5 (Nov. 2017), pp. 659–674. (4) Mary Ellen Foster and Ronald P. A. Petrick. "Separating representation, reasoning, and implementation for interaction management: Lessons from automated planning". In: Dialogues with Social Robots: Enablements, Analyses, and Evaluation. Ed. by Kristiina Jokinen and Graham Wilcock. Lecture Notes in Electrical Engineering 427. Springer, 2017, pp. 93–107. (5) Mary Ellen Foster etal, . "The MuMMER project: Engaging human-robot interaction in real-world public spaces". In: Proceedings of the Eighth International Conference on Social Robotics (ICSR 2016).

Partner Organisation: Amrita Vishwa Vidyapeetham, AU, India	
General Description	Amrita Vishwa Vidyapeetham is a multi-campus, multi-disciplinary research academia that is accredited **'A'** by NAAC and is ranked as one of the best research institutions in India. Amrita is spread across six campuses in three states of India - Kerala, Tamil Nadu and Karnataka, with the headquarters at Ettimadai, Coimbatore, Tamil Nadu. Amrita Vishwa Vidyapeetham continuously collaborates with top US universities including Ivy league universities and top European universities for regular student exchange programs, and has emerged as one of the fastest growing institutions of higher learning in India. The institution is managed by the Mata Amritanandamayi Math. Amrita Vishwa Vidyapeetham established under section 3 of UGC Act of 1956 offers more than 200 professional programs through 15 schools across 6 different campuses in 3 South Indian states. The world renowned humanitarian and spiritual leader Sri Mata Amritanandamayi Devi (Amma), known worldwide for Her humanitarian activities and initiatives, is the Chancellor. Amma, a great inspiration for the faculty, staff and student community founded the institution with a vision to impart high quality education enriched with human values to the younger generation for the benefit of society. Amrita Vishwa Vidyapeetham is Top Private Indian University according to QS BRICS University Rankings, a ranking of the top universities in the five BRICS countries (Brazil, Russia, India, China and South Africa). Amrita Vishwa Vidyapeetham is No. 1 in International Faculty according to QS BRICS University Rankings. Amrita Vishwa Vidyapeetham personifies multi-disciplinary academic excellence through the exceptional quality of its education system that is modern and comprehensive, with special emphasis on core human values. Amrita Vishwa Vidyapeetham was ranked 8th in India University Rankings 2019. AMMACHI Labs at AU brings an interdisciplinary approach to **addressing societal challenges** and as a center of excellence, we envision a world where technology is designed and used to empower people. We create innovative educational tools and skill development solutions to help **uplift rural communities**. They are engaged in **technology interventions** and community development activities in over 142 villages across 21 states in India.
Academic organisation	Yes
Role and profile of key persons (supervisor)	**Prof. Bhavani Rao** is Director of AMMACHI Labs, she also serve as the **UNESCO Chair** in Gender Equality & Women's Empowerment and head the R&D activities that focuses on Human-Computer Interaction (HCI), Robotics and Automation technologies for promoting gender equality, quality STEM education, skill development, disaster response, medical assistance and rehabilitation. She is the Principal Investigator for several projects at Amrita, funded by organisations such as the United Nations Democracy Fund (UNDEF), UNICEF, Internet.org, NORRAG, the Ministry of Human Resources & Development (MHRD) Govt of India, and the National Skill Development Corporation (NSDC). She has led or collaborated in the acquisition of over £2.8M of research funding. Prof. Bhavani and AMMACHI labs are developing expertise in assembling interdisciplinary teams, making social scientists to effectively working together rural communities.
Dept.	AMMACHI Labs
Key research facilities, Infrastructure and Equipment	I will be provided with a fully equipped working station (with PC, printer, free access to Internet, e-mail account, telephone) with an own desk located in the main University campus. I will also have free access to AU's libraries; free access to various academic courses; web-profile on AU's website; access to online scientific journals; and venues for meetings/seminars/lectures. I will have access to UGLA's digital resources from AU campus.
Independent research premises?	Yes
Previous and current involvement in research and training programmes	TO BE ADDED XXXXXXXXXXXXXXXXXXXX XXXXXXXXXXXXXXXXXXX XXXXXXXXXXXXXXXXXXX XXXXXXXXXXXXXXXXXXX XXXXXXXXXXXXXXXXXXXXXXXXXXX XXXXXXXXXXXXXXXXXXXXXXXXXXX XXXXXXXXXXXXXXXXXXXX

6 Ethical Issues

With reference to the Ethic issues table (section 4 of Part A of ENDEAVOUR), the following text is going to address the following issues: Humans, Personal data, Third countries.

ENDEAVOUR will conduct work in a manner which **complies with ethical principles**, and which adheres with existing practices carried out by AU and UGLA. Specifically, I aim to: **(1)** obtain verbal / written consent from each participant, making it clear that they can withdraw from the study or from answering particular questions; **(2)** work with school authorities; and **(3)** obtain verbal/written consent for video recordings.

1) Interviews, questionnaires: Prior to any interview and questionnaires, I will inform the research participants, in a language they understand (e.g. Hindi / Marathi / Malayalam), about the features and aims of ENDEAVOUR and I will explicitly ask for their **informed consent**. For qualitative data collection I anticipate to secure verbal informed consent, while for quantitative data collection written consent will be used (but depending on the level of literacy, verbal consent is preferred to respect participants' conditions). The verbal consent will be formally documented and independently witnessed. It will be made clear that participation is not compulsory, and participants may decline to participate in this study. The students' data including questionnaire, interview responses will **anonymised** no data will have information which can identify an individual, participants will be refereed with random ID numbers. The same applies to semi-structured interviews and questionnaires collected from teachers on their feedback about the intervention. **No sensitive personal data** will be collected throughout the life-cycle of ENDEAVOUR.

Minors: ENDEAVOUR will involve participants who will be children aged between 5-11. I will obtain written (or verbal, depending on the level of literacy) informed consent of their parents / carers / relatives / teachers/ tutors/ legal representatives. If required / appropriate, to ensure their welfare, they will be interviewed in the presence of any of the adults above as appropriate, and, in case of girls, I will always be accompanied by a female translator. The children will be informed that if they choose, they could leave the study/interview and go back to their classrooms. ENDEAVOUR will reassure them that **no coercion** will ever be exerted it will be made clear to that participation is be entirely voluntary. Children under the age of 5 will never be interviewed or be part of any studies. There will not be any issue with physical harm to the participants, as the robot will be secured on a wall near water taps and the children will not be able reach the robot. There are no moving parts on the robot except for the small eyes which cannot cause any physical harm. However, there will be someone from the research team supervising the environment all the time in case they have to intervene.

2) Schools: Informed verbal/written consent will obtained from the school authorities including the principal and the teachers who will be the guardians of the children, but also the children's parents. Informed consent will obtained from the children. All data will be collected within the confines of the school. Locations for the interviews will be intentionally chosen to be outside classroom and official environments. Prior to starting the research meetings will be arranged with the school authorities to introduce the project, explain the purpose of the research, and **seek permission** to work in the school. The reasons for conducting the work will be explained in local language. Researchers members from AU already have good rapport with the schools authorities due to their prior work so there should not any issue of trust.

Also during studies ENDEAVOUR will ensure that **no disruption** to normal school operation will happen during school hours. Interventions and interviews will be planned in the afternoon sessions during breaks. The students in the school have access to the government's scheme for free, nutritious lunches, which is a motivation for many underprivileged families to send their children to schools. In addition to this initiative, there is also a free breakfast initiative for tribal children. A typical school day starts at 10 A.M. with prayer and classes run till 12:30 P.M when the lunch break begins. In between they have a short break at 11:30 A.M. The lunch break lasts until 2:00 P.M and the classes proceed till 3:30 P.M. The intervention with the social robot will only take place between breaks to ensure not to disturb the normal school hours. Soap and water will be made available for the participants to wash their hands, which is a prerequisite for ENDEAVOUR.

3) Photos and videos: In order to guarantee the privacy rights of the research participants, I will ask a further informed and written consent for the use of photos and interviews' videos in all ENDEAVOUR-related initiatives and materials will be obtained from their parents / carers / relatives / teachers/ tutors/ legal representatives. I will **respect the subject's privacy** requirements whenever they ask me to. For data brought from India to UK, I will use automated software to pixelate faces (using Adobe After Effects) and remove audio on recorded footage after analysis has been performed.

The data generated or synthesised from ENDEAVOUR will comprise questionnaires in excel files, image and video data from field studies. All data will be made anonymous, conforming to privacy and ethical approval by AU in India and University of Glasgow ethics committee. Data will be stored **securely with regular backup** and appropriate security practices on an encrypted hard drive. Data generated will be compiled into password protected databases and files, which will be managed initially by me. Password protected access to the databases and files will be provided also to the research teams at AU and the University of Glasgow (UGLA). All data storage and provision will be **compliant** with GDPR procedures as implemented at UGLA. Data will be stored for a maximum period of five years following the project.

Although there are **no particular safety concerns** related to my work in India, normal safety precautions in terms of working in a foreign country and conducting field work will be taken. I will complete a full risk assessment using templates well developed by UGLA scientists conducting research in developing countries. Appropriate travel health insurance and required Pre-Exposure vaccinations will be procured, and when in the field, I will: work accompanied by at least one other person; ensure that others know the travel plans and itinerary; carry authorised identification; and carry a mobile phone. Originally being brought up in India I speak fluent Hindi (official language in India) and Marathi, local language in Maharashtra state.

This being a **Global Fellowship**, the ongoing phase of this project will be carried out in UGLA, UK. The fieldwork phase of ENDEAVOUR will take place in a non-EU country, India. Since this research will be studying the interaction between rural subjects (children) and robots, this work **could not be performed elsewhere**. Given that the research directly involves people and their data **no sensitive personal identifiable data**, clearance will be obtained through the College Ethics Committee of the universities of Glasgow (UK) and Amrita University (India). Application procedures, which will be explicitly related to ENDEAVOUR, will include: submission of the research proposal (including copies of data collection protocols and participant information sheets and consent forms for those involved in interviews, study and focus groups) and institutional ethical clearance. Two months will be allowed for this process before initiation of the field phase of the research. At the end of this period and before the start of the field studies ENDEAVOUR, a scanned copy of ethics approval documents (in their original form in English) will be provided to the REA. I will **not proceed** with any research which requires ethical approval before the REA has received the documents listed below:

1. Information Sheets in English / Hindi / Malayalam / Marathi which:
   (a) describe the aims, methods and implications of ENDEAVOUR, the nature of the participation and any benefits, risks or discomfort that might be involved
   (b) explicitly state that participation is voluntary and that anyone has the right to refuse to participate and to withdraw their participation or data at any time — without any consequences
   (c) indicate how data will be collected, protected during the project and either destroyed or reused subsequently
2. Informed Consent Forms in English / Hindi / Malayalam / Marathi to ask permission for the collection, storing, and use of video, questionnaire data
   (a) Copies of Ethics Approvals (UGLA, AU)
   (b) Copy of notifications / authorisations for the collection and processing of data from parents / carers / relatives / teachers/ tutors/ legal representatives and school authorities
   (c) Copy of notifications / authorisations for data transfer to UK from India
   (d) Confirmation that the research is compatible with EU and international law

Humans	
I confirm that templates of the informed consent forms and information sheets (in language and terms intelligible to the participants) will be kept on file.	Yes
Third country	
I confirm that the research performed outside the EU is compatible with the Union, National and International legislation and could have been legally conducted in one of the EU Member States.	Yes
Data protection	
I confirm that a Data Protection Officer (DPO) has been appointed and the contact details of the DPO are made available to all data subjects involved in the research.	Yes
I confirm that data intended to be processed is relevant and limited to the purposes of the research project (in accordance with the 'data minimisation' principle).	Yes

7 Letter of Commitment

ammachi labs

Amrita Multi Modal Applications Using Computer Human Interaction Labs
Amrita Vishwa Vidyapeetham
Amritapuri

Phone: +91-476 2804409
Fax: +91-476 2804493
ammachilabs@amrita.edu
www.amrita.edu/ammachi

To,
Research Executive Agency MSCA-IF
European Commission
Brussels

Dated: 11th September 2019

Dear Sir/Madam,

Subject: Letter of support for Dr. Amol Deshmukh's application to the Marie Sklodowska-Curie Individual Global Fellowship 2019

On behalf of AMMACHI Labs at Amrita University, India we are pleased to write this letter in support of Dr. Amol Deshmukh's application entitled "Influencing Handwashing Behaviour Using Social Robots" (ENDEAVOUR), submitted under Sklodowska-Curie Individual Global Fellowship program.

In the proposed project, Dr. Deshmukh will be investigating the influence of handwashing behavior using social robots with rural subjects in India along with researchers at AMMACHI Labs. Apart from being the Director of AMMACHI Labs, I also serve as the UNESCO Chair in Gender Equality & Women's Empowerment and head the R&D activities that focus on Human-Computer Interaction (HCI), Technology Enhanced Learning and Robotics and Automation technologies for skill development, women empowerment, and gender equality. I am the Principal Investigator for several projects at Amrita, funded by organizations such as the United Nations Democracy Fund (UNDEF), Internet.org, NORRAG, the Ministry of Human Resources & Development (MHRD) and the National Skill Development Corporation (NSDC). AMMACHI Labs brings an interdisciplinary approach to addressing societal challenges and as a Centre of Excellence, we envision a world where technology is designed and used to empower people. We create innovative educational tools and skill development solutions to help uplift rural communities. We are engaged in technology interventions and community development activities in over 142 villages in India, and we believe ENDEAVOUR will contribute and extend our research activities. Our group is also involved with rural health programs to spread handwashing awareness and promotion where we have trained hundreds of children and adults in rural areas in safe handwashing practices. We also have an extensive network of social workers working with schools and after school tuition centers in rural areas actively working with over 57 schools in 21 states with the potential to work with more than a thousand children, which is ideal for ENDEAVOUR.

Dr. Deshmukh's proposal is timely and extremely relevant for the Indian context as India has a low 15% handwashing prevalence rate. His innovative idea of using low-cost social robots for influencing handwashing behavior is relevant to the national campaign launched by the Indian government "Swachh Bharat – Swachh Vidyalaya" (Clean India – Clean Schools)

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Amrita Vishwa Vidyapeetham
Amritapuri

Phone: +91-476 2804409
Fax: +91-476 2804493
ammachilabs@amrita.edu
www.amrita.edu/ammachi

working to improve water, sanitation, and hygiene (WASH) services and practices across India. To the best of our knowledge, there is no research relevant to using social robots for such applications in rural populations in India, hence his research idea has the potential to create a positive impact on the wider community. Dr. Deshmukh will get training on the design of qualitative and quantitative methodologies for carrying out field studies in rural villages supplemented with researchers from the AMMCHI labs team. As the primary subjects of this research will be children, Dr. Sidney Strauss, an expert in child development and education and Dr. Meltem Akoyak-Yildiz from pedagogic and education sciences from Amrita University will provide guidance for his research.

We were pleased to have Dr. Deshmukh as a visiting researcher in 2017 to study the technology perception of rural subjects with a use case of water carrying social robot which resulted in two publications and global media publicity. Recently he also led a team of researchers from AMMACHI labs in March 2019 to carry out a pilot study with a social robot to influence handwashing behavior in a rural school in Kerala, India and the study has resulted again in two publications. His fellowship proposal is very much in line with our strategic vision and we would be able to provide support for his field studies and enable him to carry out this project in the most effective way possible.

Its is my pleasure to serve as Dr. Deshmukh's supervisor and host at Amrita University for the India based portion of his fellowship and liaise closely with the overall project supervisor, Dr. Mary Ellen Foster at Glasgow University. In our role of providing host arrangements during Dr. Deshmukh's time at Amrita, he will join our research team at AMMACHI labs as a visiting fellow and have access to our lab, office spaces, computing resources, makerspace and library resources. Dr. Deshmukh will also get the opportunity to connect with other resources and labs across Amrita University and participate in seminars, give talks on his work as well as public engagement activities in a variety of settings.

We look forward to Dr. Deshmukh's receipt of this prestigious fellowship and the opportunity to welcome him to Amrita University. Please contact me if I can provide any additional information.

Yours sincerely,

Prof. Bhavani Rao
Director, AMMACHI Labs
UNESCO Chair and Director,
Center for Women Empowerment and Gender Equality
Amrita Vishwa Vidyapeetham University,
Amritapuri Campus, Kollam, Kerala, India.
Phone: +91 9447498820
Email: bhavani@ammachilabs.org

AMRITA VISHWA VIDYAPEETHAM AMRITAPURI CAMPUS KOLLAM